

NEWPORT, AR, USA

WMO: 720377

Lat: 35.633N

Lon: 91.167W

Elev: 0

StdP: 14.70

Time Zone: -6.00 (NAC)

Period: 06-19

WBAN: 13917

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 17.1	(c) 20.7	(d) 4.8	(e) 7.0	(f) 21.1	(g) 9.6	(h) 9.0	(i) 24.7	(j) 28.4	(k) 46.7	(l) 26.2	(m) 48.0	(n) 9.8	(o) 350	(p) 0.491

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 18.6	(c) 98.3	(d) 78.4	(e) 95.4	(f) 78.0	(g) 92.9	(h) 77.2	(i) 81.6	(j) 91.4	(k) 80.5	(l) 90.4	(m) 79.4	(n) 89.5	(o) 10.0	(p) 170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.3	152.0	86.0	78.2	146.4	85.3	76.7	139.1	84.3	45.2	91.4	43.9	90.7	42.8	89.1	85.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.7	22.4	19.4	11.3	101.0	4.5	3.1	8.0	103.2	5.4	105.0	2.9	106.7	-0.4	109.0
			9.7	83.6	4.1	1.1	6.8	84.4	4.4	85.0	2.0	85.7	-0.9	86.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	61.0	37.9	41.7	51.7	60.9	70.8	80.3	81.4	79.9	73.2	61.8	49.6	42.1
	DBStd	17.51	10.29	11.06	10.58	8.47	7.38	4.89	4.61	4.95	6.16	8.74	8.97	9.11
	HDD50	1198	394	279	108	15	1	0	0	0	0	9	114	278
	HDD65	3482	840	654	426	173	36	0	0	0	8	170	466	709
	CDD50	5230	19	46	160	343	645	909	974	927	697	374	102	34
	CDD65	2038	0	1	13	50	215	459	509	462	255	70	4	0
	CDH74	20855	0	7	91	386	1821	5161	5612	4811	2259	677	30	0
	CDH80	9454	0	0	8	67	648	2560	2686	2310	956	216	3	0
(14) Wind	WSAvg	8.8	10.0	10.5	10.9	10.9	9.1	8.1	7.0	6.3	6.6	7.9	8.8	9.3
(15) Precipitation	PrecAvg	48.8	4.4	3.7	4.7	4.5	4.7	3.6	4.1	2.7	3.3	4.0	4.3	4.7
	PrecMax	73.7	8.2	7.8	9.9	11.5	10.4	6.1	8.2	4.4	8.1	12.2	9.1	10.2
	PrecMin	33.1	1.7	1.1	1.3	0.4	0.3	1.1	1.1	0.3	0.0	0.2	0.9	0.3
	PrecStd	9.0	1.7	1.8	2.1	2.3	2.4	1.5	2.0	1.3	1.9	2.8	2.4	2.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	67.9	74.0	81.2	84.5	92.7	99.3	99.8	101.6	94.8	90.2	78.8	70.4
		MCWB	63.0	64.3	65.7	69.2	73.4	77.5	78.6	80.1	76.3	72.6	65.8	65.4
	2%	DB	63.2	69.1	76.5	81.9	89.9	97.1	97.3	98.6	92.4	85.9	72.6	65.6
		MCWB	58.8	61.8	65.4	67.3	72.9	77.3	78.8	79.1	76.2	70.7	62.8	61.7
	5%	DB	58.2	64.1	72.3	79.0	86.8	94.1	94.7	94.2	90.0	81.6	68.4	61.2
		MCWB	52.8	58.3	62.2	65.9	71.8	76.3	78.9	78.5	75.6	67.7	61.3	57.3
	10%	DB	54.2	59.6	67.7	75.0	83.7	91.4	91.9	91.0	86.1	77.1	64.0	56.5
		MCWB	49.3	54.8	58.9	64.3	71.1	75.5	78.6	78.0	73.6	66.0	57.8	52.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	63.8	66.6	69.1	73.0	77.2	81.3	83.1	83.5	80.0	76.1	69.2	66.7
		MCDB	66.2	71.4	77.9	80.4	87.7	93.3	92.3	94.4	89.7	85.4	74.6	69.0
	2%	WB	60.3	63.5	66.5	70.2	75.5	79.8	81.7	81.8	78.0	73.7	66.0	62.8
		MCDB	62.8	67.0	74.0	78.3	84.1	91.9	91.0	91.8	88.8	82.2	70.4	64.8
	5%	WB	54.7	59.8	64.1	68.4	74.3	78.5	80.6	80.5	76.7	71.2	62.6	58.0
		MCDB	56.8	63.7	69.8	75.3	82.6	89.9	90.2	90.2	86.6	77.8	65.9	60.5
	10%	WB	49.8	55.1	61.8	66.5	72.9	77.5	79.6	79.2	75.3	68.5	59.1	53.7
		MCDB	53.8	58.3	66.8	72.6	80.7	88.4	89.6	88.6	83.3	74.0	63.6	56.1
(35) Mean Daily Temperature Range	5% DB	MDBR	16.7	17.2	19.1	20.0	18.8	19.9	18.6	19.5	21.1	22.7	19.8	15.6
		MCDBR	22.8	24.0	23.5	23.3	22.2	22.5	21.6	23.5	24.3	27.2	24.8	20.8
		MCWBR	17.6	17.7	14.0	11.6	8.8	6.9	6.8	8.2	9.9	13.0	15.6	16.6
	5% WB	MCDBR	19.4	20.5	21.1	19.4	18.6	20.1	19.1	20.3	21.2	22.5	20.0	18.6
		MCWBR	17.7	17.0	13.7	11.9	8.7	7.5	7.6	9.0	9.4	12.7	15.7	16.7
(40) Clear-Sky Solar Irradiance	taub		0.324	0.330	0.366	0.408	0.446	0.473	0.498	0.487	0.423	0.362	0.333	0.317
	taud		2.464	2.448	2.378	2.254	2.214	2.152	2.121	2.130	2.300	2.452	2.486	2.500
	Ebn at Noon		277	289	286	278	268	260	252	252	264	274	272	273
	Edh at Noon		27	31	36	43	45	48	49	48	39	30	26	25
(44) All-Sky Solar Radiation	RadAvg		769	943	1302	1649	1826	2081	1987	1828	1550	1189	839	661
(45)	RadStd		65	125	104	117	134	134	94	117	117	146	89	72

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (25 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page